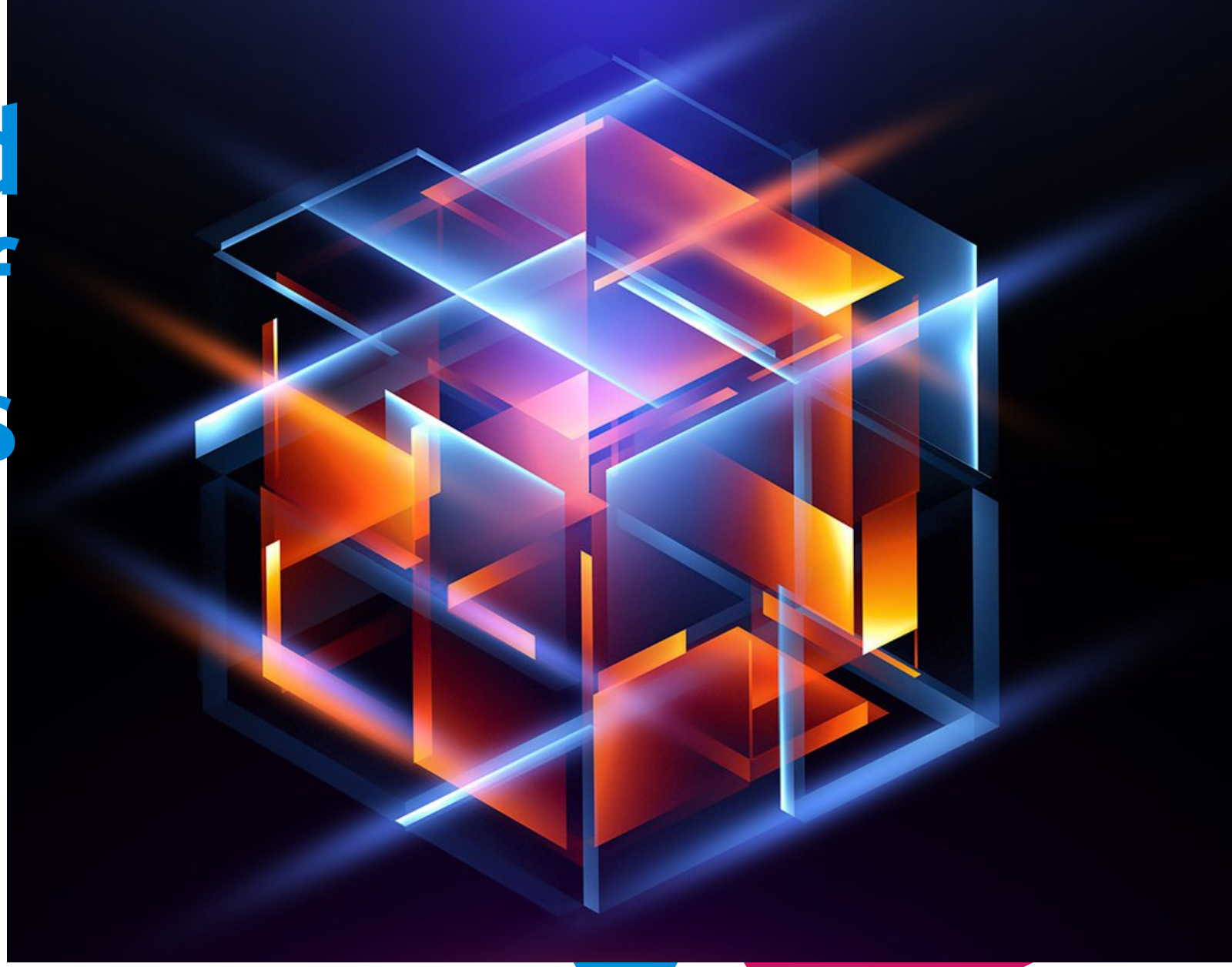
Desing and Analysis of Algorithms

Introduction



What is an Algorithm?

Motivation for Complexity Analysis

Big-O Notation

- Expressing running time in terms of *basic computer steps.*
- *Simplification of running time* as a function of the size of the input.
- And this simplification leads to another.
  - Instead of reporting, $5n^3 + 4n + 3$ steps on an input of size n.
  - $4n$ and 3 become insignificant as n grows.
  - even the detail of the coefficient 5.
    - computers will be five times faster in a few years anyway.
  - Then, the algorithm takes time $O(n^3)$.

Example - Dominant term method

```
int sumArray(int arr[], int n) {
    int sum = 0;
    for (int i = 0; i < n; i++)
        sum += arr[i];
    return sum;
}
```

Example - Dominant term method

Line	Operations	Count
`int sum = 0;`	1 assignment	1
`for (int i = 0; i < n; i++)`	`i=0` (1), `i<n` (n+1), `i++` (n)	1 + (n + 1) + n = 2n + 2
`sum += arr[i];`	addition + assignment	2n
`return sum;`	1 return	1

✅ **Total operations:**

$$1 + (2n + 2) + 2n + 1 = 4n + 4$$

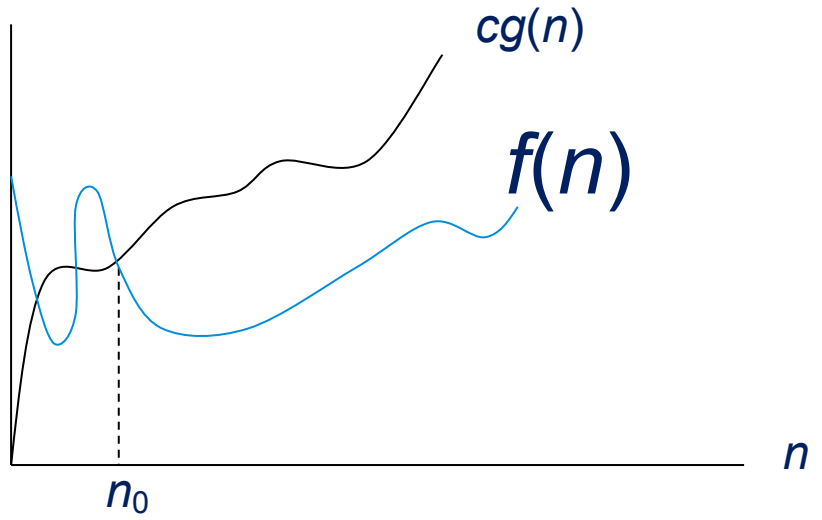
- **Exact number of operations:** $4n + 4$
- **Asymptotic complexity (Big-O):** $O(n)$

Big-O Notation

- Think of $f(n)$ and $g(n)$ as the running times of two algorithms on inputs of size n.
- Let $f(n)$, $g(n)$: N ⇒ R^+
- We say $f = O(g)$ (which means that f grows no faster than g) if *there is a constant $c > 0$ an n_0 such that*

$$f(n) \leq c\ g(n)$$

for all $n \geq n_0$



$f(n) = O(g(n))$

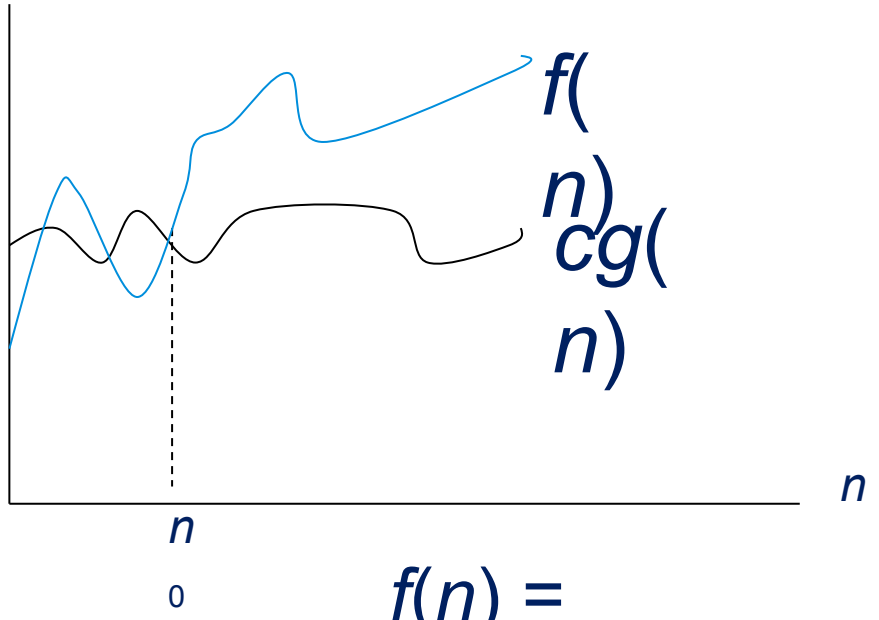
Lemma If $\lim_{n\to\infty} \frac{f(n)}{g(n)} = L$, where $0 \leq L < \infty$, then $f(n) = O(g(n))$.

Ω-Notation

- We say $f = \Omega(g)$ (which means that f grows no better than g) if *there is a constant $c > 0$ an n_0 such that*

$$f(n) \geq c\ g(n)$$

for all $n \geq n_0$



$f(n) = \Omega(g(n))$

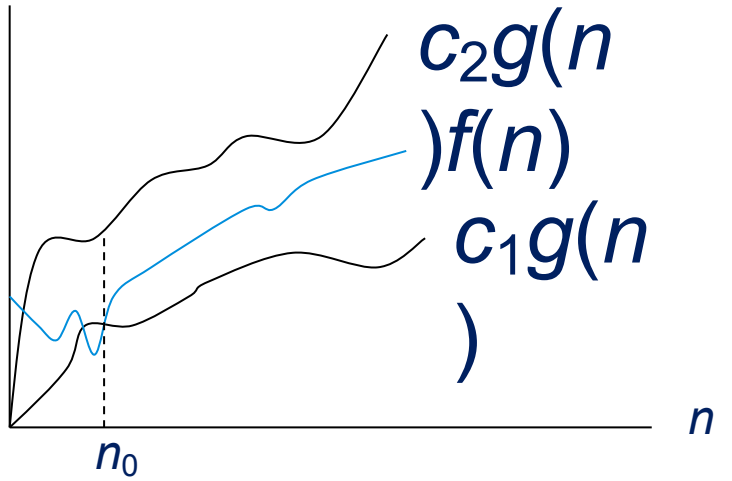
Lemma If $\lim_{n\to\infty} \frac{f(n)}{g(n)} = L$, where $0 < L \leq \infty$, then $f(n) = \Omega(g(n))$.

Θ-Notation

- We say $f = \Theta(g)$ (which means that f approximately behaves like g) if *there two constants* $c_1, c_2 > 0$ *an* n_0 *such that*

$$c_1\ g(n) \le f(n) \le\ c_2\ g(n)$$

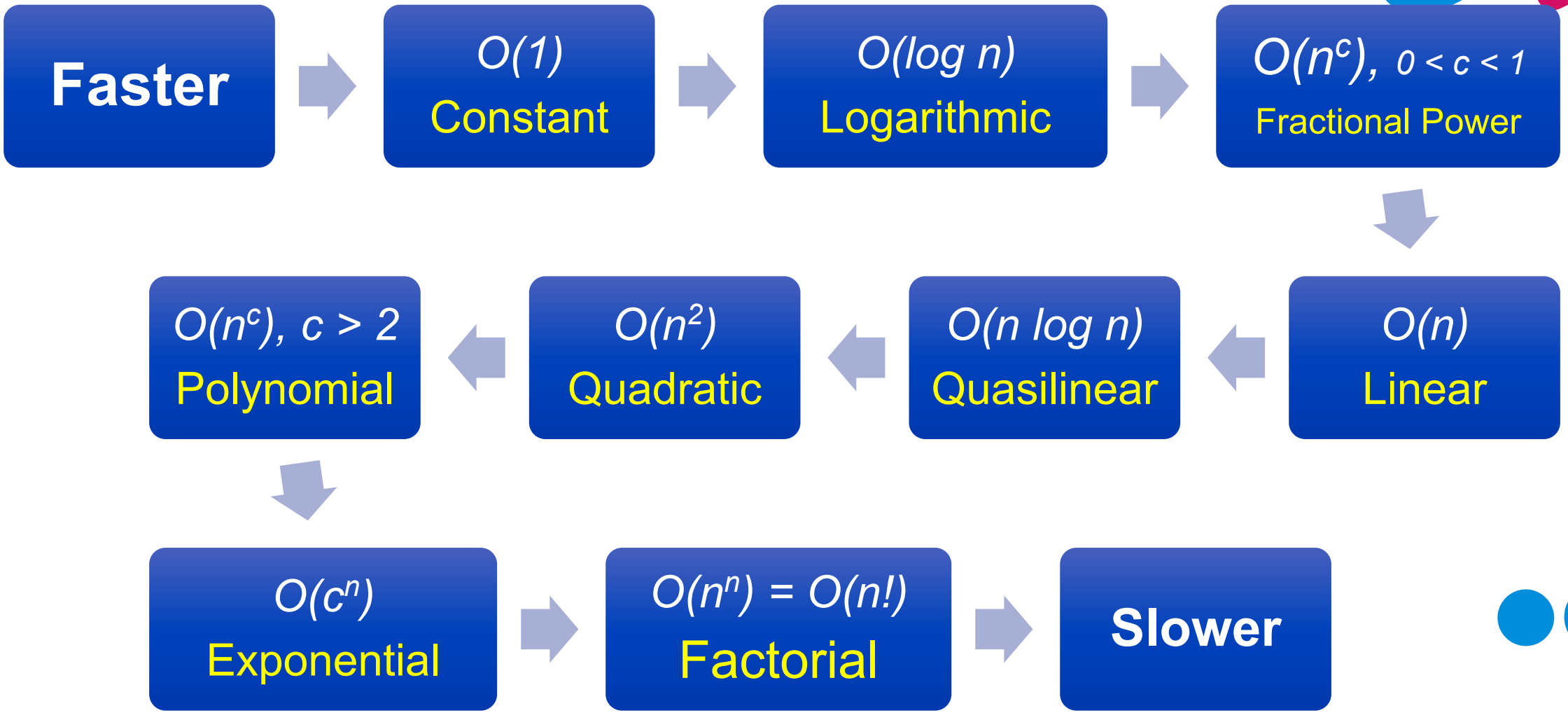
for all $n \ge n_0$



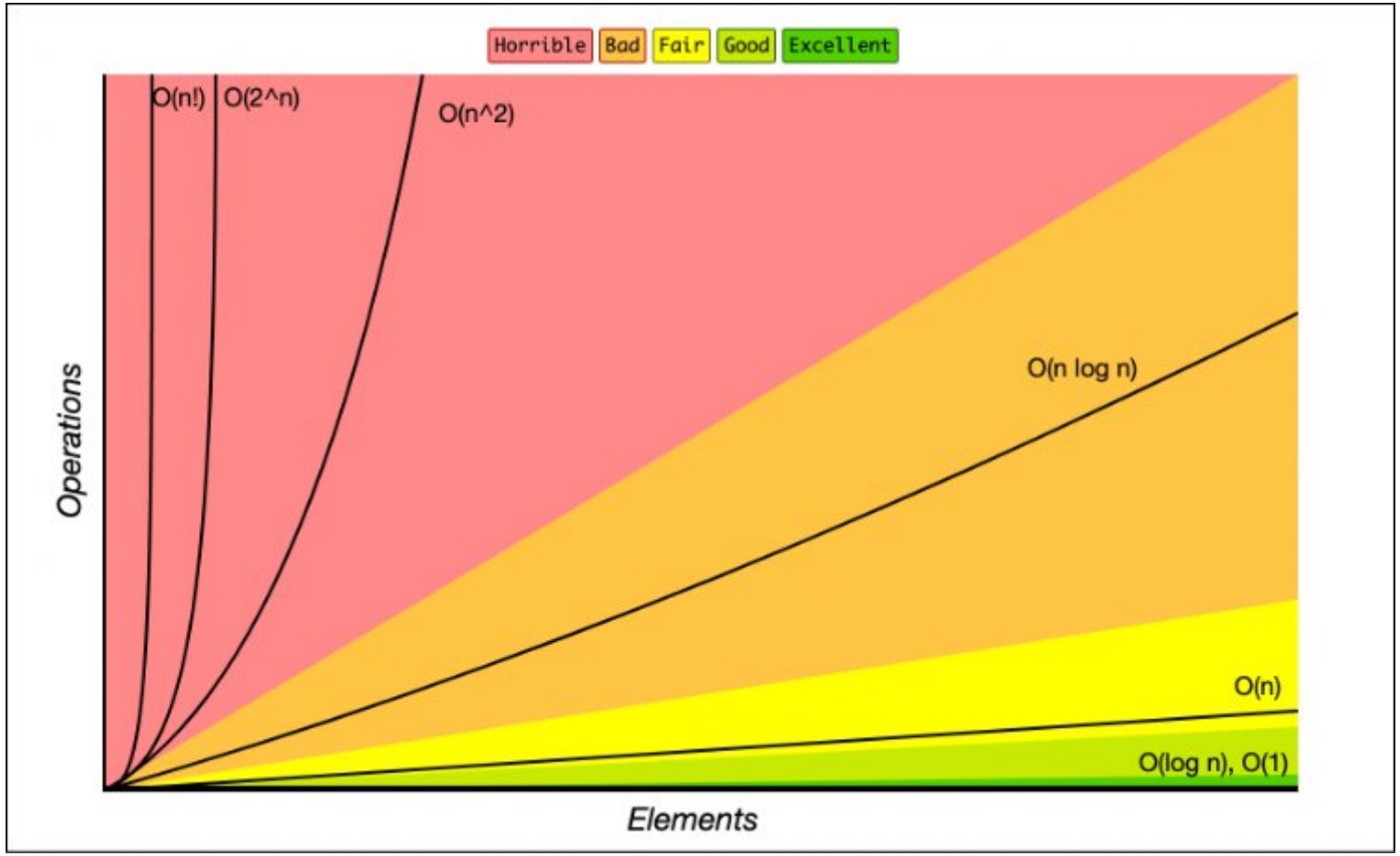
$f(n) = \Theta(g(n))$

Lemma If $\lim_{n\to\infty} \frac{f(n)}{g(n)} = L$, where $0 < L < \infty$, then $f(n) = \Theta(g(n))$.

Order of Growth



Order of Growth



Example - Using Limits

Proof (using limits)

We want to show $f(n) = 3n^2 + 5n + 7$ is $\Theta(g(n))$ for $g(n) = n^2$. A standard limit test: if

$$\lim_{n\to\infty} \frac{f(n)}{g(n)} = L$$

with $0 < L < \infty$, then $f(n) = \Theta(g(n))$.

Compute the ratio and its limit:

$$\frac{f(n)}{g(n)} = \frac{3n^2 + 5n + 7}{n^2} = 3 + \frac{5}{n} + \frac{7}{n^2}.$$

Now take the limit term-by-term as $n \to \infty$:

- $\lim_{n\to\infty} 3 = 3.$
- $\lim_{n\to\infty} \frac{5}{n} = 0$ (since dividing a fixed number by $n \to \infty$ tends to 0).
- $\lim_{n\to\infty} \frac{7}{n^2} = 0.$

Therefore

$$\lim_{n\to\infty} \frac{f(n)}{g(n)} = 3 + 0 + 0 = 3.$$

Since 3 is a finite positive constant, the limit test gives $f(n) = \Theta(g(n))$.

Grades

Questions

